ZARAGOZA. LOGISTICS CENTER

MZ 505
Inventory Systems

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Session 19: Risk Pooling II

MIT GLOBAL SCALE NETWORK

Product pooling

- The pooling benefits of serving the same aggregate demand with less product variety.

Surf Hammer 3/2 logo

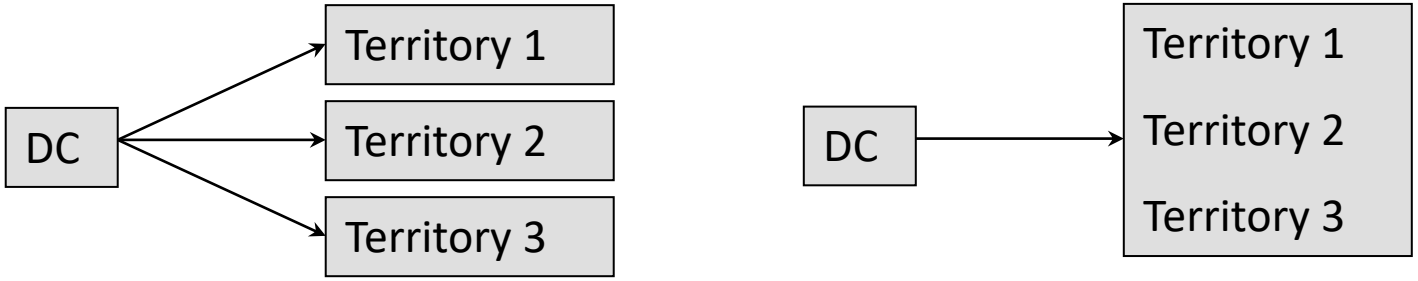
Dive Hammer 3/2 logo



- Example: Instead of having two Hammer 3/2 suits, O'Neill could consolidate its product line into a single Hammer 3/2 suit, i.e., a universal design, which we will call the "Universal Hammer".
- A key force behind the "SKU rationalization".

Location pooling

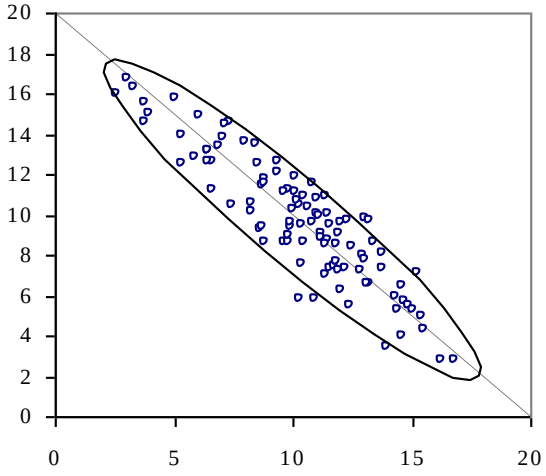
- The pooling benefits of serving the same aggregate demand from fewer locations.

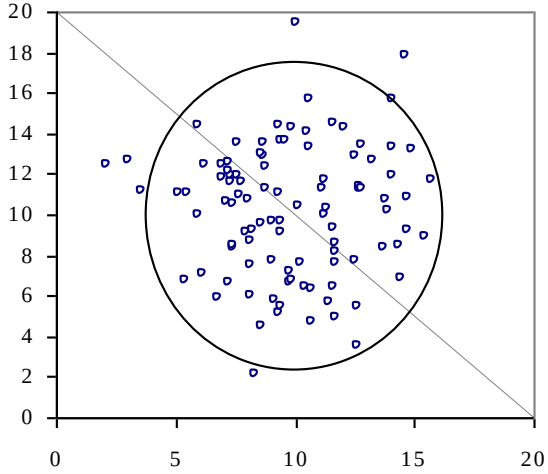


- Demand variability is reduced because as demand is aggregated across different locations, it becomes more likely that high demand from one customer will be offset by low demand from another.

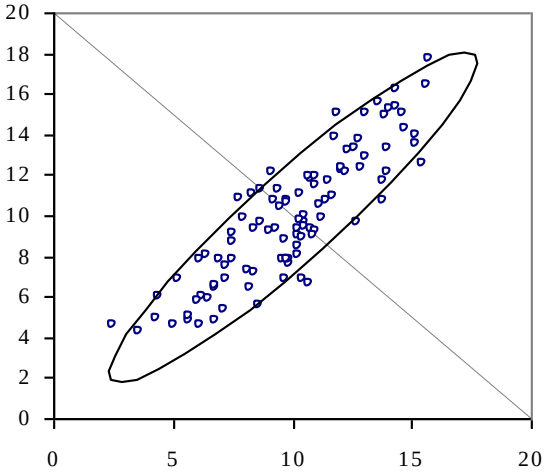
Correlation

Correlation: quantitative measure of the strength and direction of the relationship between two random variables.





Random demand for two products (x-axis is product 1, y-axis is product 2). In scenario 1 (upper left graph) the correlation coefficient is -0.9, in scenario 2 (upper right graph) the correlation is 0 and in scenario 3 (the lower graph) the correlation is 0.9. In all scenarios demand is Normally distributed for each product with mean 10 and standard deviation 3.



Covariance and Correlation

- The covariance of two random variables X and Y, denoted by cov(X, Y) is defined by

$$\mathrm{cov}(X,Y) = E\big[(X - E[X])(Y - E(Y))\big] = E[XY] - E[X]E[Y]$$

- The correlation coefficient ρ(X, Y) of two random variables X and Y that have nonzero variances is defined by

$$\rho(X,Y) = \frac{\mathrm{cov}(X,Y)}{\sqrt{\mathrm{var}(X)\mathrm{var}(Y)}} \quad \rightarrow \quad \rho \in [-1,1]$$

- The variance of the sum of two random variables X and Y is

$$\mathrm{var}(X+Y) = \mathrm{var}(X) + \mathrm{var}(Y) + 2\mathrm{cov}(X,Y)$$

- More generally, for n random variables $X_1, X_2, \ldots, X_n$:

$$\mathrm{var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{var}(X_i) + \sum_{\{(i,j)|i\neq j\}} \mathrm{cov}(X_i, X_j)$$

Demand forecast and risk pooling

- The three principles of all forecasting techniques:
  - Forecasts are always wrong.
  - The longer the forecast horizon the less accurate is the forecast.
  - Aggregate forecasts are always more accurate than individual forecasts.
    - ✓ **The Risk Pooling Concept**

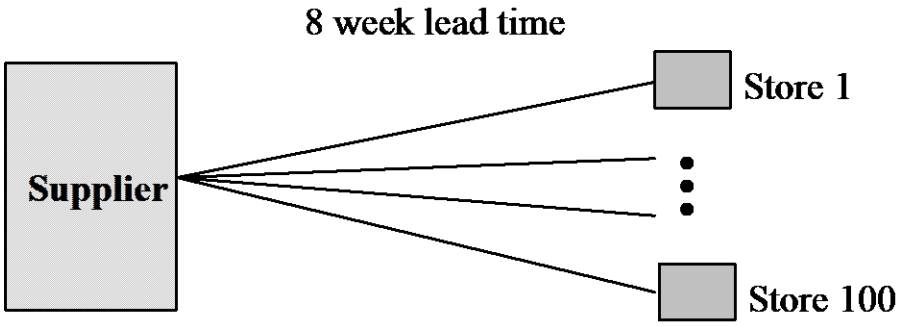


Lead Time Pooling: Consolidated Distribution

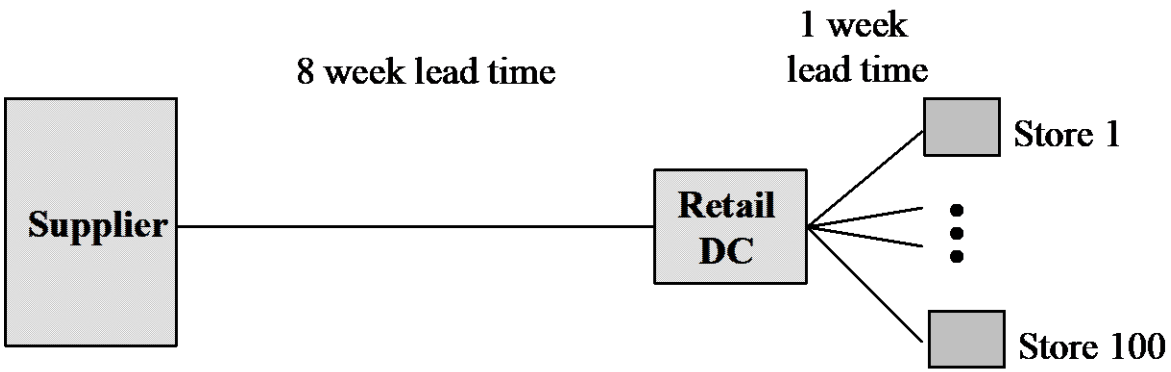
Example: Consolidated distribution

- Consider the following supply chain:
  - The weekly demand at each store is Poisson with mean 0.5 and the target in-stock probability at each store is 99.5%.

Current system: direct from supplier



Proposed system: centralized inventory in a distribution center



- DC weekly demand is normally distributed with mean 50 and standard deviation 15
- If demands were independent across stores, then DC demand would have a standard deviation of sqrt(50) = 7.07

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System inventory with no pooling

Unconsolidated system

- Assume periodic review with one week review period.
- Demand over L+1 periods is Poisson with mean (8+1)×0.5 = 4.5
- Choose S = 11 to satisfy 99.5% in-stock probability.
- Inventory at each store:

$$E[\text{on-hand}] = S - \mu_{L+1} + E[\text{back}]$$

$$\approx 11 - 4.5 = 6.5$$

- Total safety stock at the stores

= 6.5×100 = 650

Mean demand = 4.5		
S	**F(S)**	**L(S)**
0	0.0111	4.500
1	0.0610	3.511
2	0.1735	2.572
3	0.3422	1.745
4	0.5321	1.088
5	0.7029	0.620
6	0.8310	0.323
7	0.9134	0.154
8	0.9597	0.067
9	0.9829	0.027
10	0.9933	0.010
11	0.9975	0.003
12	0.9991	0.001
13	0.9997	0.000

System inventory with pooling

Consolidated system

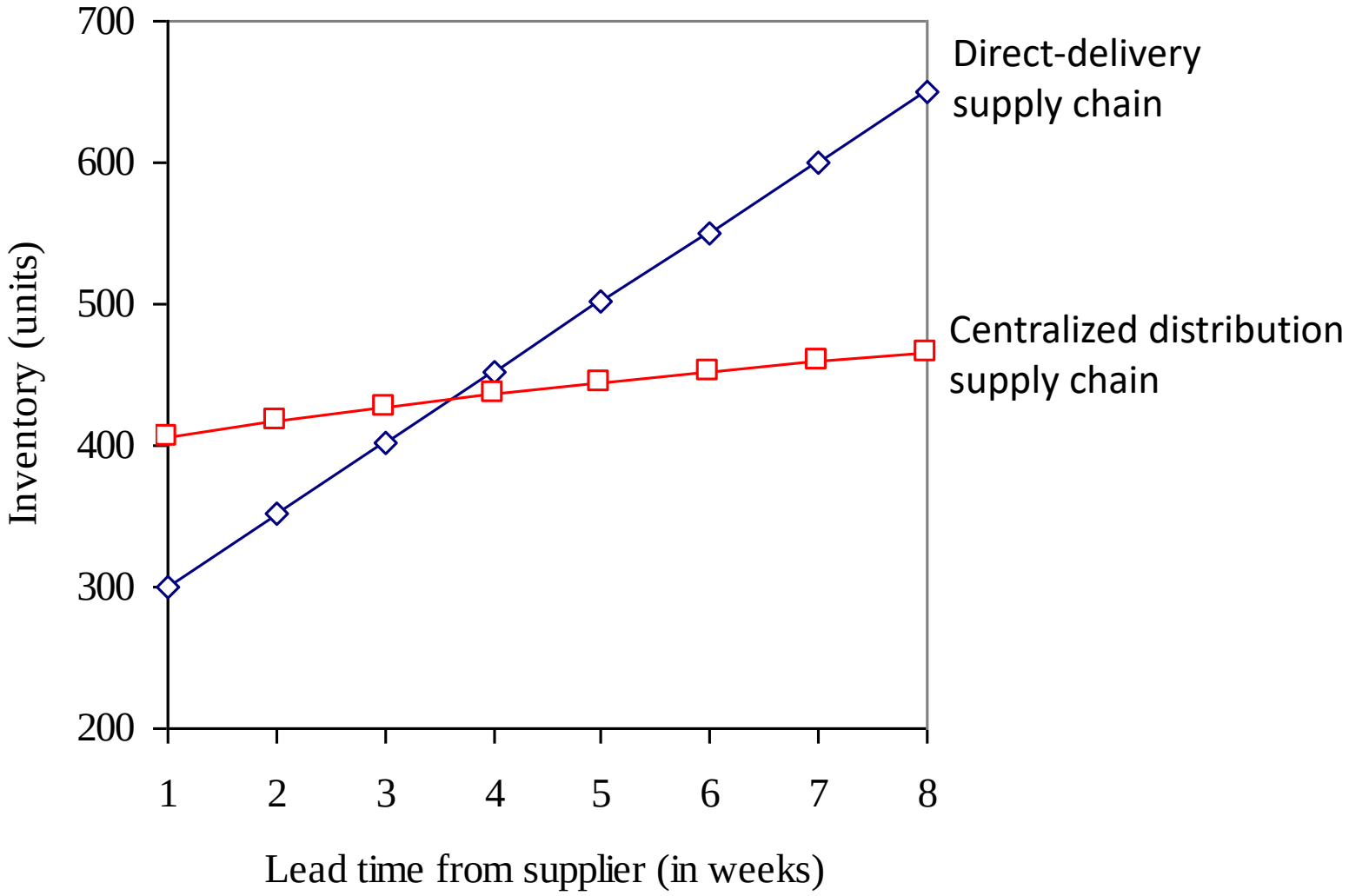
- Assume periodic review with one week review period for both the stores and the DC.
- Stores inventory
  - Demand over L+1 periods is Poisson with mean (1+1)×0.5 = 1
  - Choose S = 4 to satisfy 99.5% in-stock probability
  - For each store: E[on-hand] = S – μ_{L+1} + E[back] ≈ 4 – 1 = 3
- DC inventory
  - Demand over L+1 periods is Normal with μ_{L+1} = 450 and σ_{L+1} = 45
  - Choose S = 566 to satisfy 99.5% in-stock probability
  - For the DC: E[on-hand] = S – μ_{L+1} + E[back] ≈ 566 – 450 = 116
- Pipeline inventory between the DC and the stores = 0.5×100×1 = 50 (by Little's law)

Inventory improvement with consolidated distribution

	Direct delivery supply chain	Centralized inventory supply chain	Location pooling
Expected total inventory at the stores	650	300	0
Expected inventory at the DC	0	116	116
Pipeline inventory between the DC and the stores	0	50	0
Total	650	466	116

- Consolidated distribution …
  - Reduces retail inventory by more than **25%**!
  - Is not as effective as location pooling in reducing inventory… but keeps inventory near demand (thereby avoiding additional shipping costs to customers).
  - Reduces inventory even though the total lead time increases from 8 to 9 weeks!

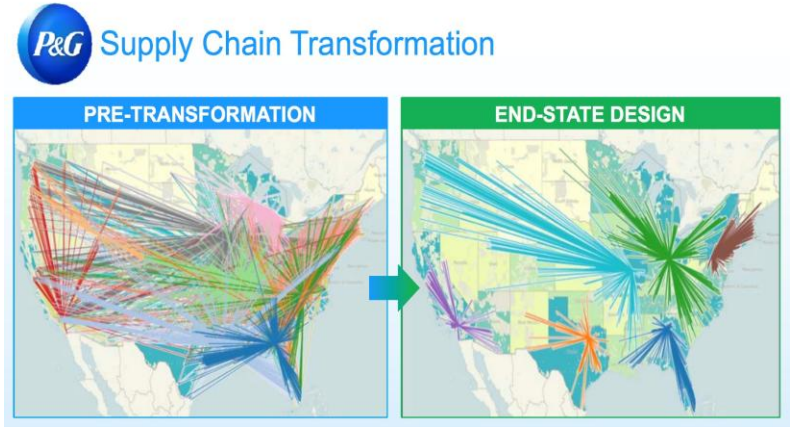
Consolidated distribution: Inventory vs. lead time



Forbes With brand divestments almost over, here's how P&G plans to cut costs next

- **Brand Consolidation Program**: After a decade of aggressive acquisitions, P&G plans to trim its portfolio of nearly 200 brands down to just 65.
- **Local Manufacturing Facilities**: P&G plans to open at least 18 new manufacturing facilities in developing markets, anticipating savings in the form of lower manufacturing, transportation and import costs.
- **Supply-Chain Optimization**: P&G anticipates 5-year savings in excess of $1billion from the consolidation of its US distribution network into fewer, larger centers, strategically located near major demand clusters.





Source: Forbes (May 2015): https://www.forbes.com/sites/greatspeculations/2015/05/07/with-brand-divestments-almost-over-heres-how-pg-plans-to-cut-costs-next/#7673f48132e2

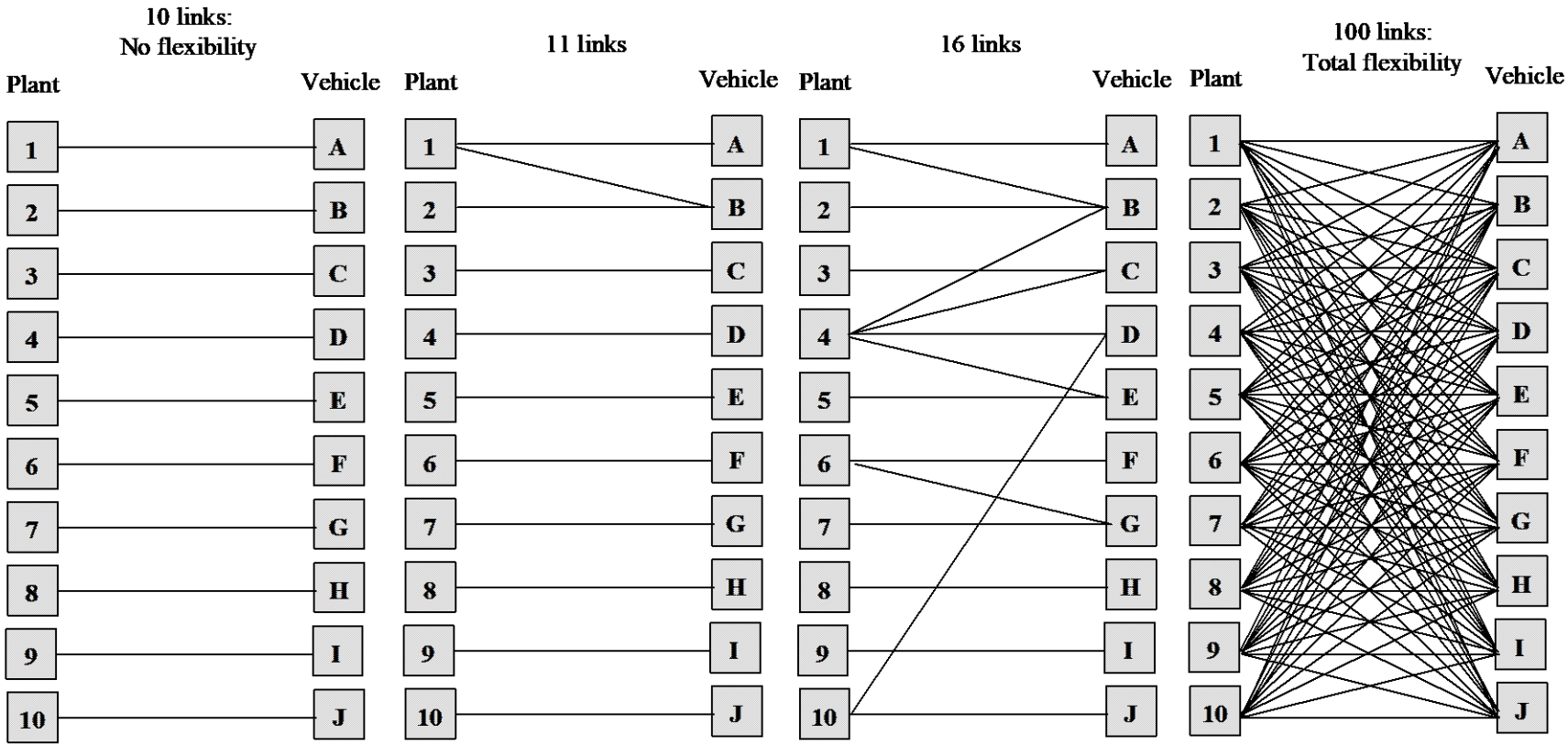
LT pooling – Consolidated distribution: Summary

- The pooling benefits of consolidating distribution closer to demand locations.
- The supply chain only needs to decide the total quantity to ship from the supplier, not its allocation across locations; hence, uncertainty is reduced.
- Advantages: Lower inventory, economies of scale in transportation and easier to obtain quantity discounts in purchasing.
- Issues to consider: Operating cost of the DC; transportation cost to stores.
- Most effective if the supplier lead time is long and the DC to store lead time is short.
- Consolidated distribution is less effective as demands across locations become more positively correlated.

Capacity Pooling

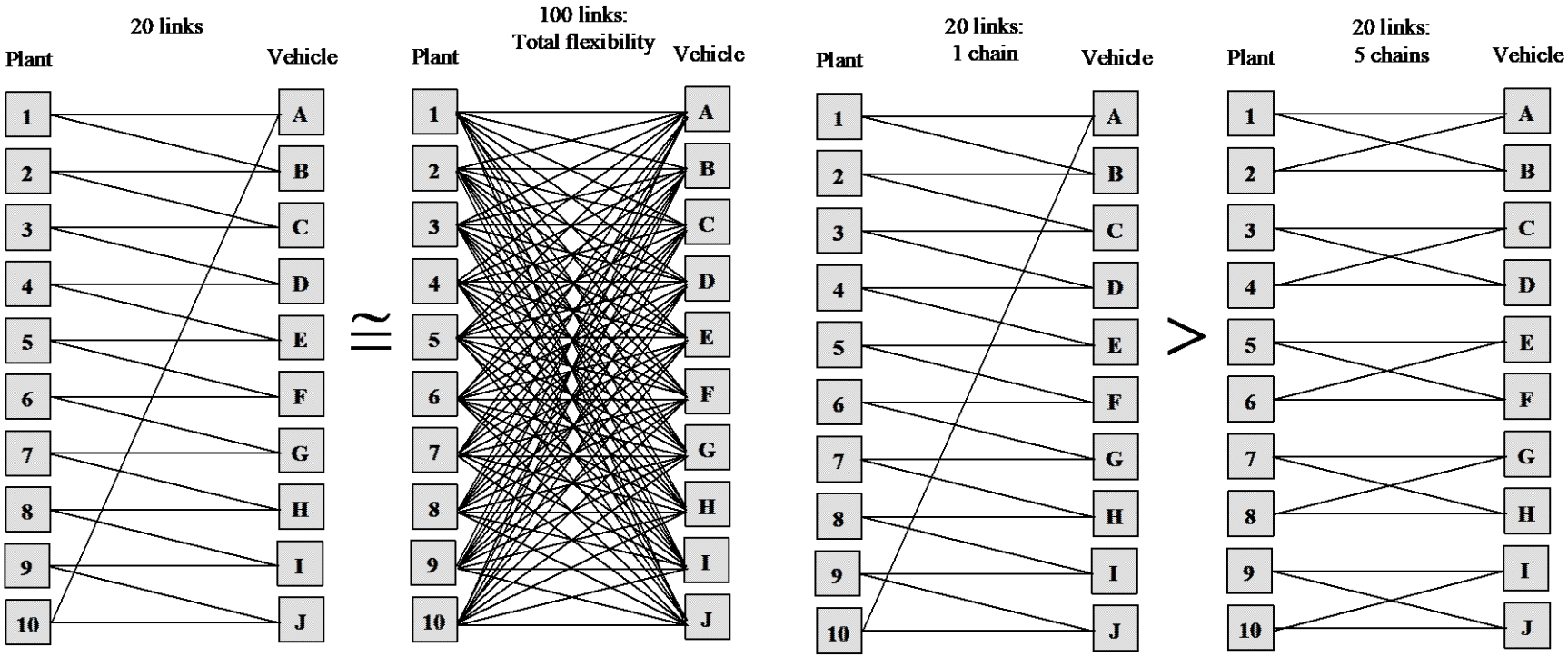
Example: Capacity pooling

- Consider the following manufacturing capacity allocation into different vehicle models.
  - The more links in the configuration, the more flexibility constructed.
  - In the 16 link configuration plant 4 is flexible enough to produce 4 products but plant 5 has no flexibility (it produces a single product).



Capacity pooling: Value of chaining

- A chain is a group of plants and products connected via links.
- Flexibility is most effective if it is added to create long chains.
- A configuration with 20 links can produce nearly the results of total flexibility as long as it constructs one large chain.
- Hence, a little bit of flexibility is very useful as long as it is designed correctly.

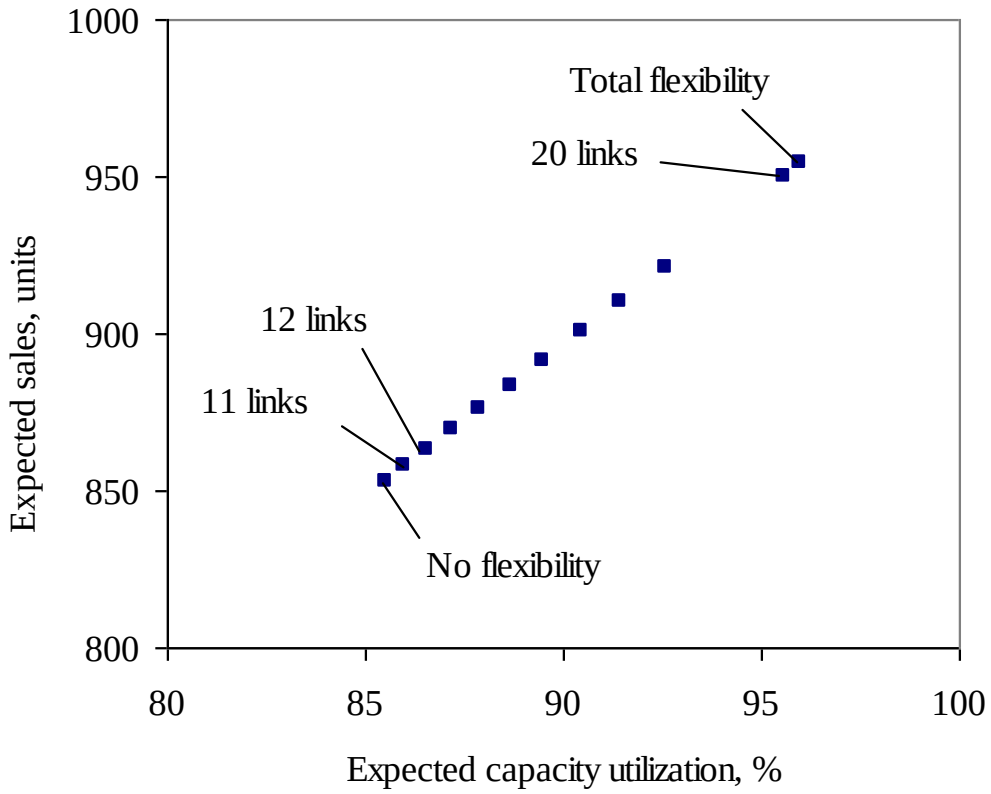


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Total flexibility vs. One long chain

Demand per vehicle Normal (μ = 100, σ = 40); Capacity = 100 / plant

- Adding flexibility increases capacity utilization and expected sales:

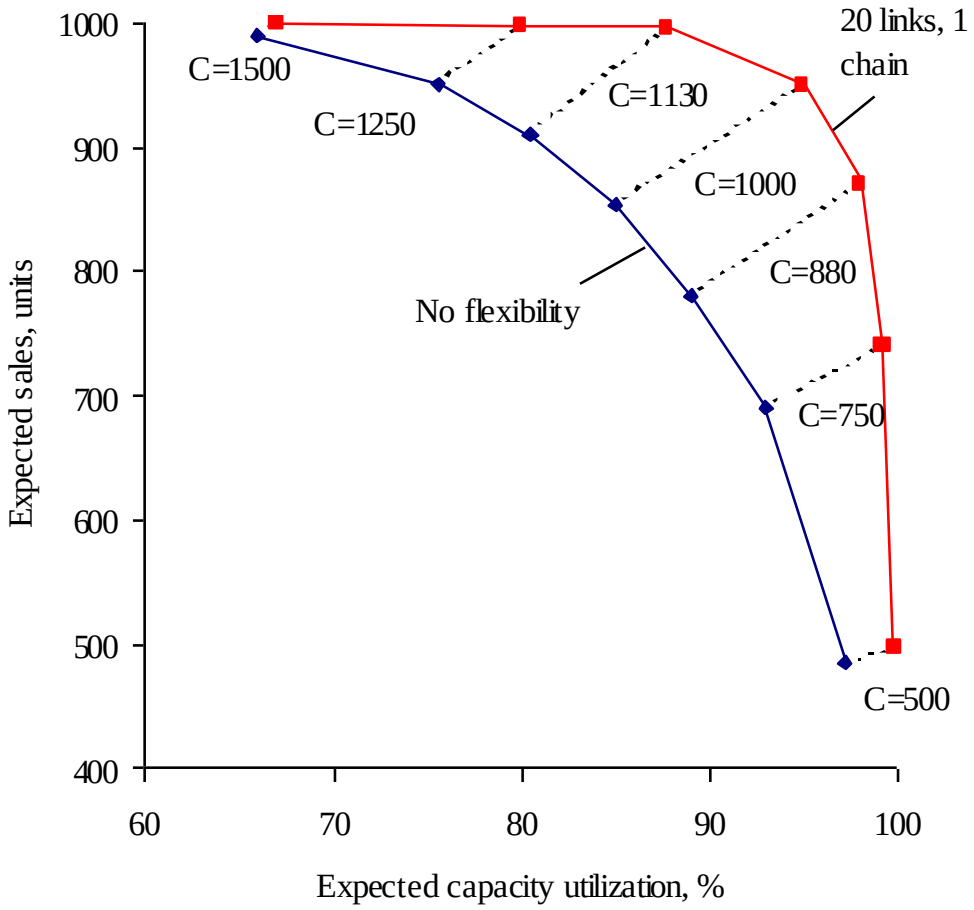


These data are collected via simulation

Note: 20 links can provide nearly the same performance as total flexibility!

Substitutability of flexibility and capacity

Demand per vehicle Normal (μ = 100, σ = 40); Total capacity (C) equally split among plants



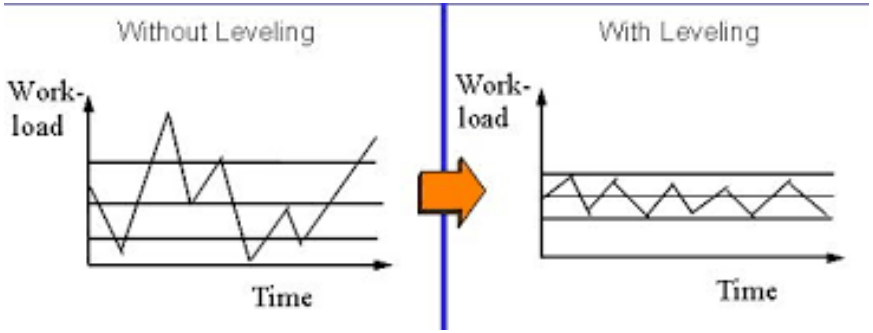
Observations:

- Flexibility is most valuable when capacity approximately equals expected demand.
- A 20 link (1 chain) configuration with 1000 units of capacity produces the same expected sales as 1250 units of capacity with no flexibility.
- If flexibility is cheap relative to capacity, add flexibility.
- But if flexibility is expensive relative to capacity, add capacity.

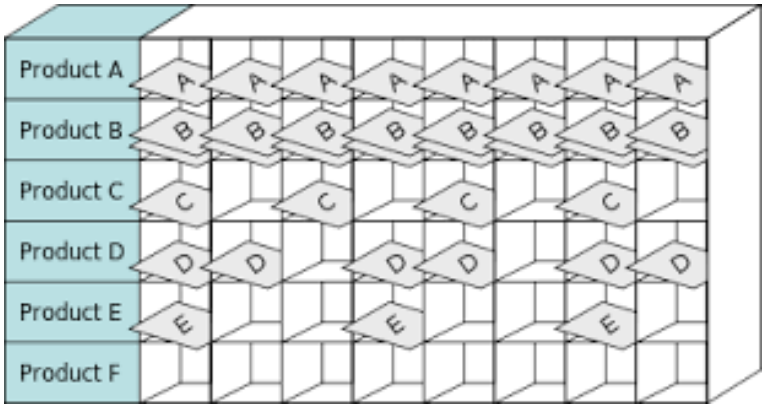
Related concept: Heijunka

- Heijunka is a Lean method for reducing the unevenness in a production process and minimizing the chance of overburden.
  - The term Heijunka comes from Japanese and literally means leveling.





Heijunka box (a scheduling tool using Kanban cards to provide a visual representation of the upcoming production runs)



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Capacity pooling: Summary

- The benefits of pooling capacity via manufacturing flexibility.
- Advantages:
  - A limited amount of flexibility can accommodate demand uncertainty nearly as well as total flexibility.
- Issues to consider:
  - Flexibility configuration (generation of long chains maximize efficiency).
  - Cost of flexibility vs. cost of capacity.
- Capacity pooling is less effective as demands across products become more positively correlated.

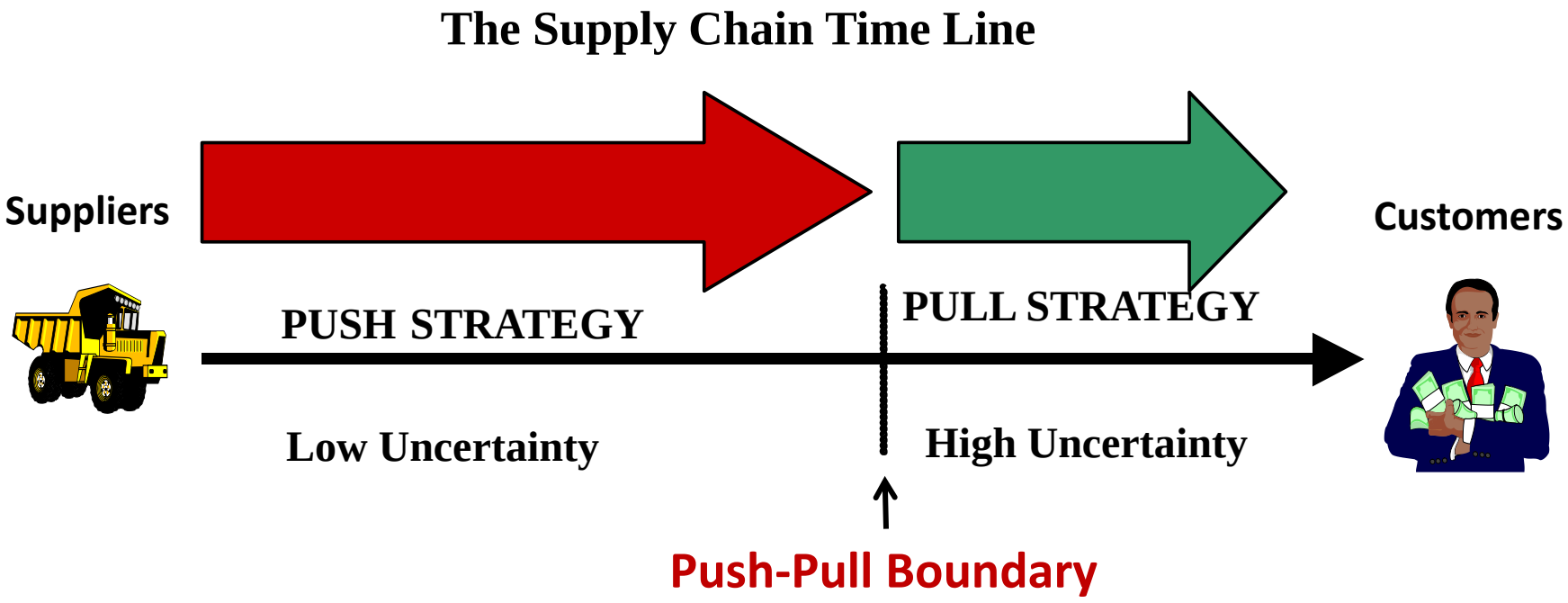
Lead Time Pooling: Delayed Differentiation

Hammer 3/2: Delayed differentiation

- A Universal Hammer 3/2 increases O'Neill's profit, but does not allow O'Neill to differentiate between the Surf and the Dive markets.
- Delayed differentiation is an alternative to the Universal Hammer:
  - O'Neill stocks "generic" Hammers with no logo.
  - When demand occurs O'Neill silk screens on the appropriate logo.
  - This generates the same profit as the Universal Hammer!
- **When does delayed differentiation make sense?**
  - Customers demand variety.
  - There is less uncertainty with total demand than demand for individual versions.
  - Variety is created late in the production process and can be added quickly and cheaply.
  - Component needed for variety is inexpensive relative to the generic component.

Delayed differentiation (or postponement)

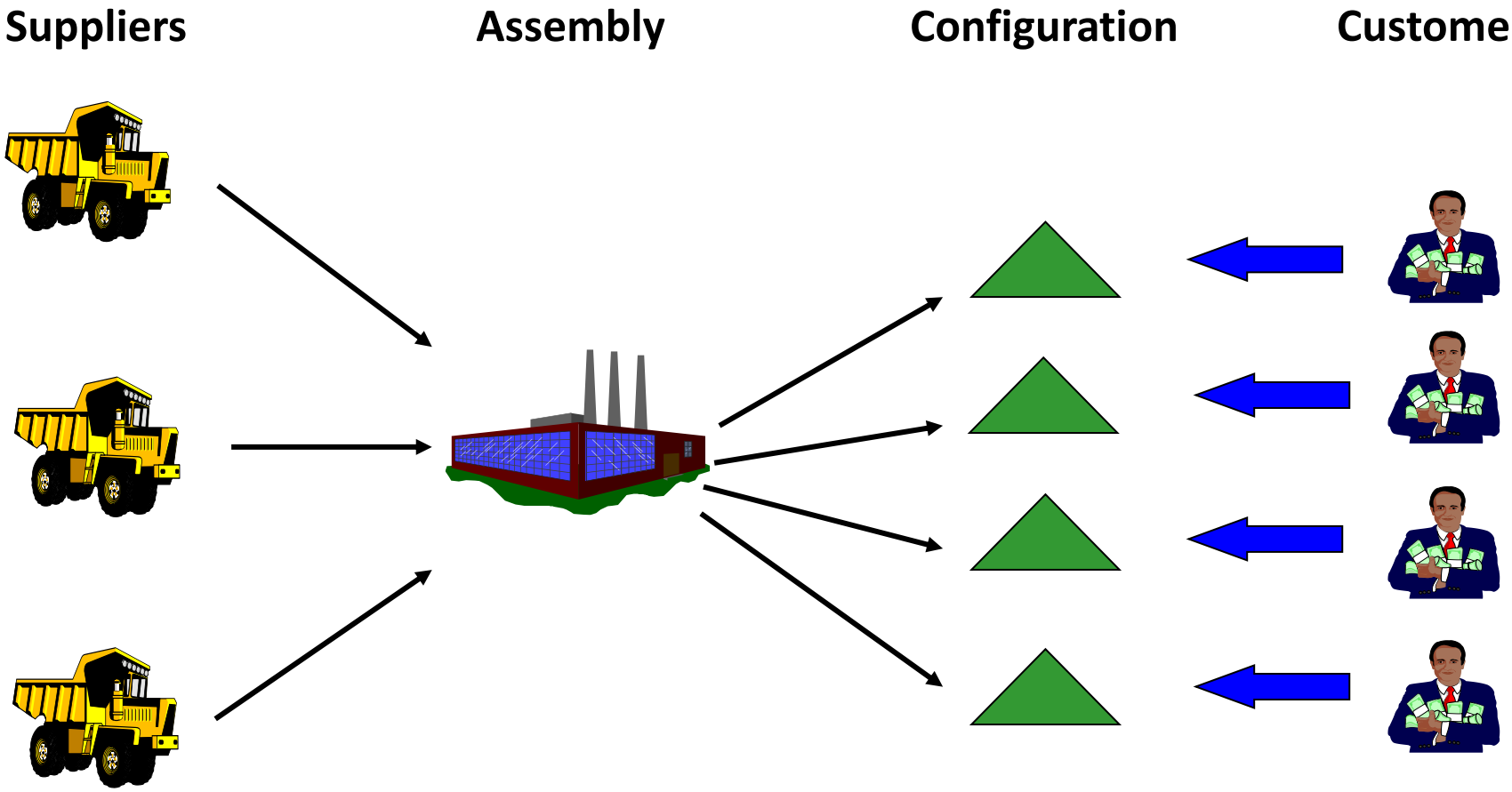
- Postponement is the ability of a supply chain to delay differentiation or customization of a product until closer to the time the product is sold.
- The goal is to have common components in the supply chain for most of the push phase and move product differentiation as close to the pull phase as possible.



From D. Simchi-Levi (2004)

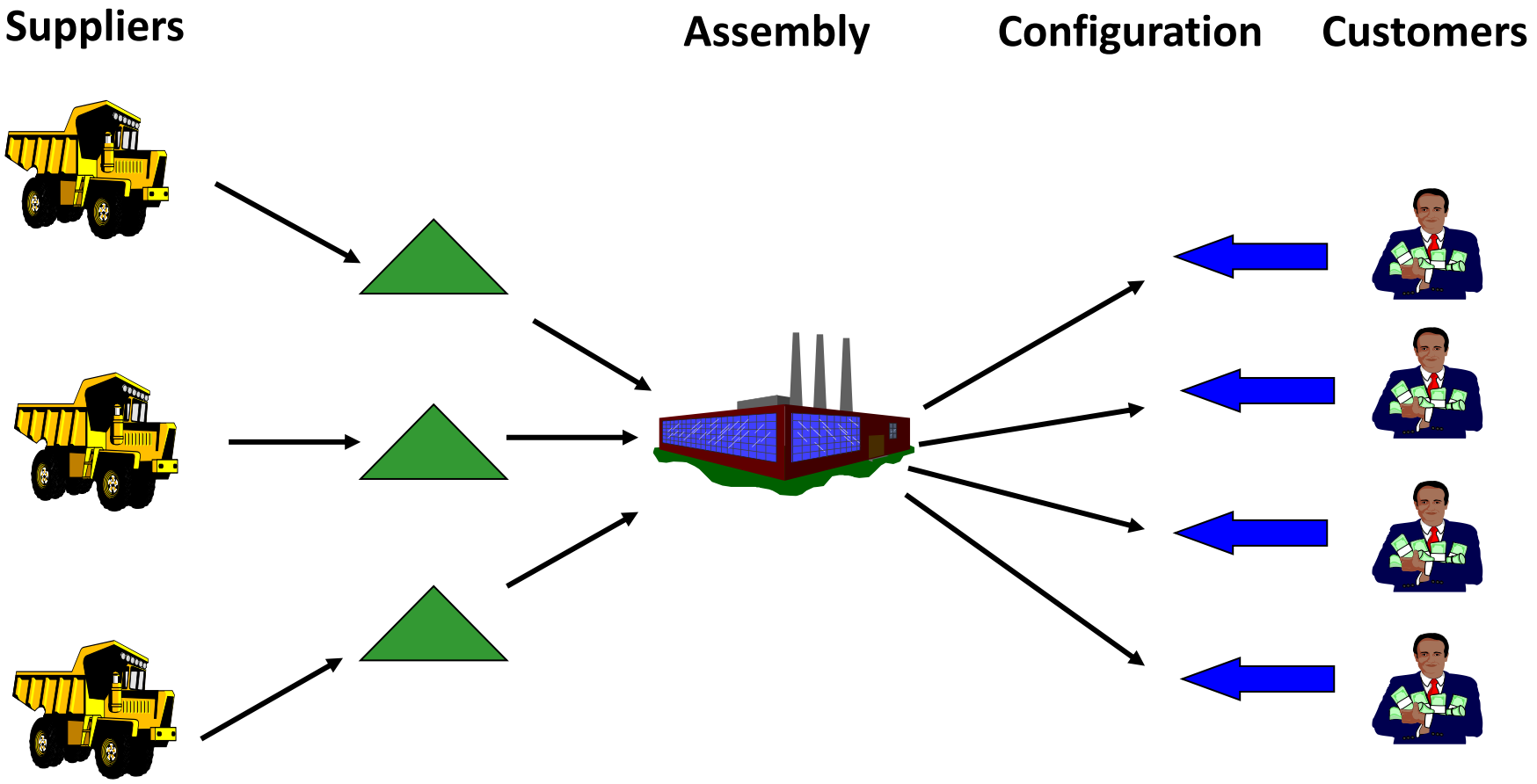
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From Make-to-Stock Model….



From D. Simchi-Levi (2004)

….to Assemble-to-Order Model

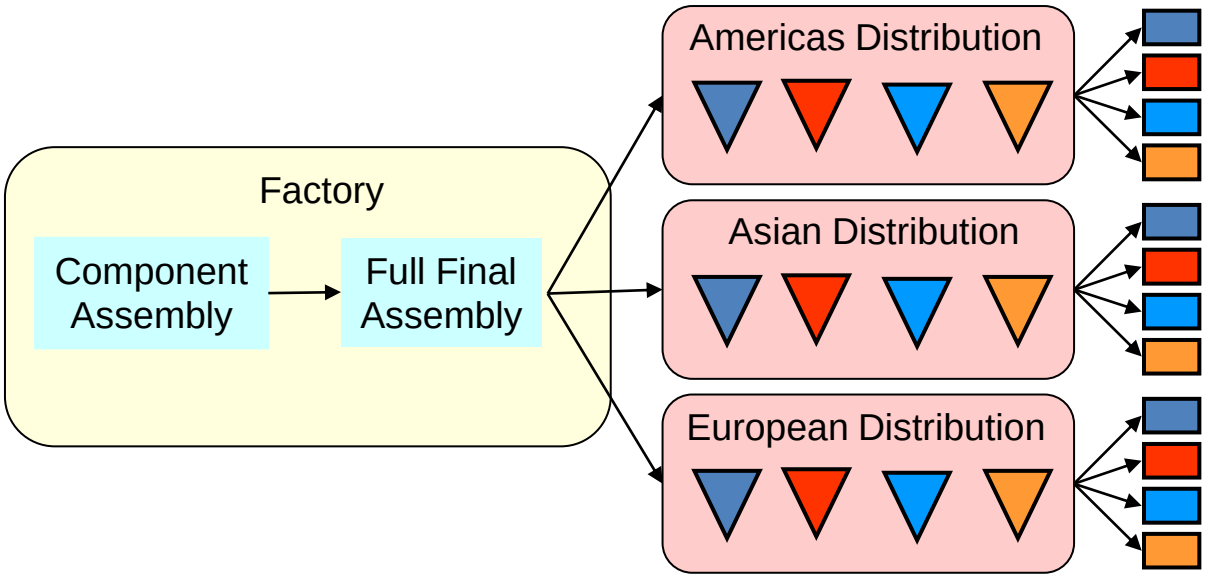


From D. Simchi-Levi (2004)

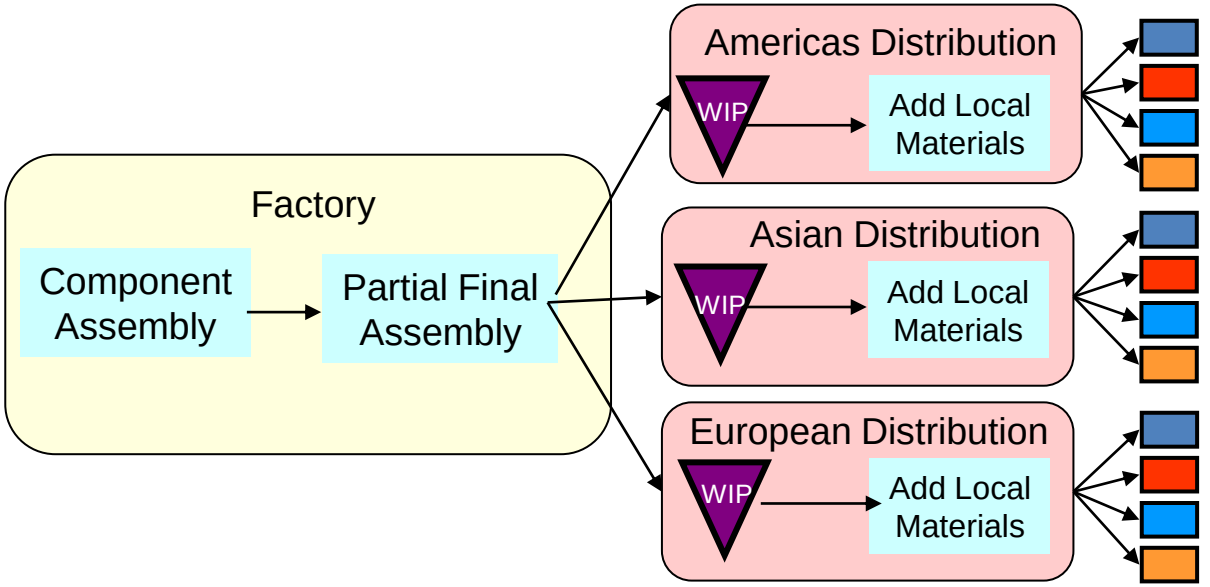
HP Printers: Customization at distributor

Before Postponement





After Postponement



Benetton sweaters

- Benetton's knitted sweaters are initially produced all white.
- They are dyed into different colors only when the season (customer) preferences are known.
- Usually the product must be specifically (re)designed for delayed differentiation.
  - This might also involve modification and/or resequencing of production processing steps.



Example: Value of postponement

- A paint retailer sells 100 different colors of paint.
- Weekly demand for each color is independent and is normally distributed with a mean of 30 and a standard deviation of 10.
- The replenishment lead time is two weeks and the retailer aims for a CSL = 0.95 (assume continuous review policy).
- How much safety stock will the retailer have to hold if paint is mixed at the factory and held in inventory at the retailer as individual colors?
- How does the safety stock requirement change if the retailer holds base paint and mixes colors on demand?

Example: Value of postponement (solution)

D =30/week; σ_D = 10; L = 2 weeks; CSL = 0.95; n=100

- Safety stock without postponement:

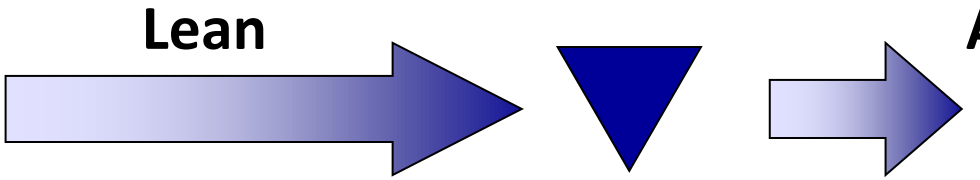
$$SS = n \cdot \Phi^{-1}(CSL) \cdot \sqrt{L} \cdot \sigma_D$$
$$= 100 \cdot \Phi^{-1}(0.95) \cdot \sqrt{2} \cdot 10 = 2{,}326$$

- Safety stock with postponement:

$$\sigma^P = \sqrt{n} \cdot \sigma_D = \sqrt{100} \cdot 10 = 100$$

$$SS = \Phi^{-1}(CSL) \cdot \sqrt{L} \cdot \sigma^P$$
$$= \Phi^{-1}(0.95) \cdot \sqrt{2} \cdot 100 = 233$$

The decoupling point



- Forecast at generic (aggregate) level
- Economic batch quantities
- Maximize efficiencies

Strategic Inventory

- Demand driven
- Localized configuration
- Maximize responsiveness

Issues to consider:

- Where to place inventory
- Economies of scale
- Product / process redesign
- Delays due to postponement stage

The effect of demand uncertainty & economies of scale

Economies of Scale	Demand Uncertainty: Low (push)	Demand Uncertainty: High (pull)
Low (pull)	**Push or pull** (Determined by cost structure & uncertainty)	**Pure "pull" strategy**
High (push)	**Pure "push" strategy**	**Push-pull** (Decouple production from distribution)

Source: Adapted from D. Simchi-Levi, *Operations Rules* (MIT Press, 2010)

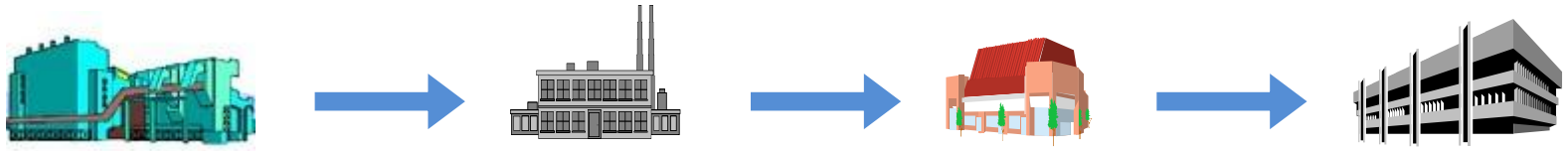
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The effect of demand uncertainty & lead time

Lead Time	Demand Uncertainty: Low (push)	Demand Uncertainty: High (pull)
Short (pull)	**Continuous replenishment**	**Pull**
Long (push)	**Push**	**Inventory positioning**

Source: Adapted from D. Simchi-Levi, *Operations Rules* (MIT Press, 2010)

Positioning of decoupling points



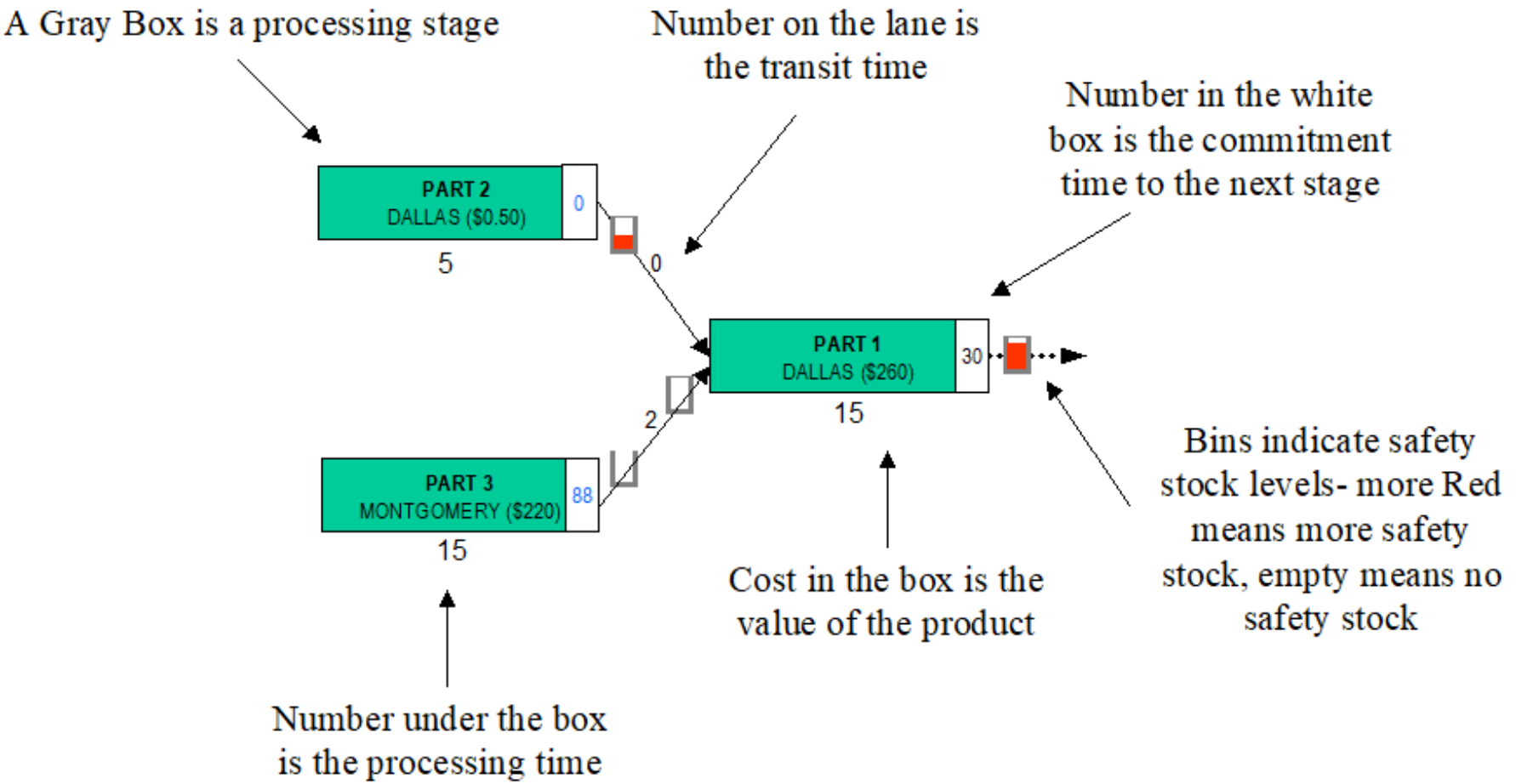
- The more product variety or the more difficult it is to predict demand, the further upstream should the decoupling point be located.
- Place decoupling buffers before high value-adding processes where inventory costs rise significantly.
- Place before "bottleneck" or highly capacity constrained processes.
- Place between processes that should operate with different levels of production smoothing.

Shifting the "push-pull" boundary: A case study

- Manufacturer of circuit boards and other high-tech products
- Sells customized products with high value and short life cycles
- Multi-stage BOM
  - e.g., copper & fiberglass ⇒ circuit board ⇒ enclosure ⇒ processor
- Case study concerns a number of the 27,000 SKUs

Source: Adapted from Simchi-Levi, Chen, & Bramel, *The Logic of Logistics* (Springer, 2004)

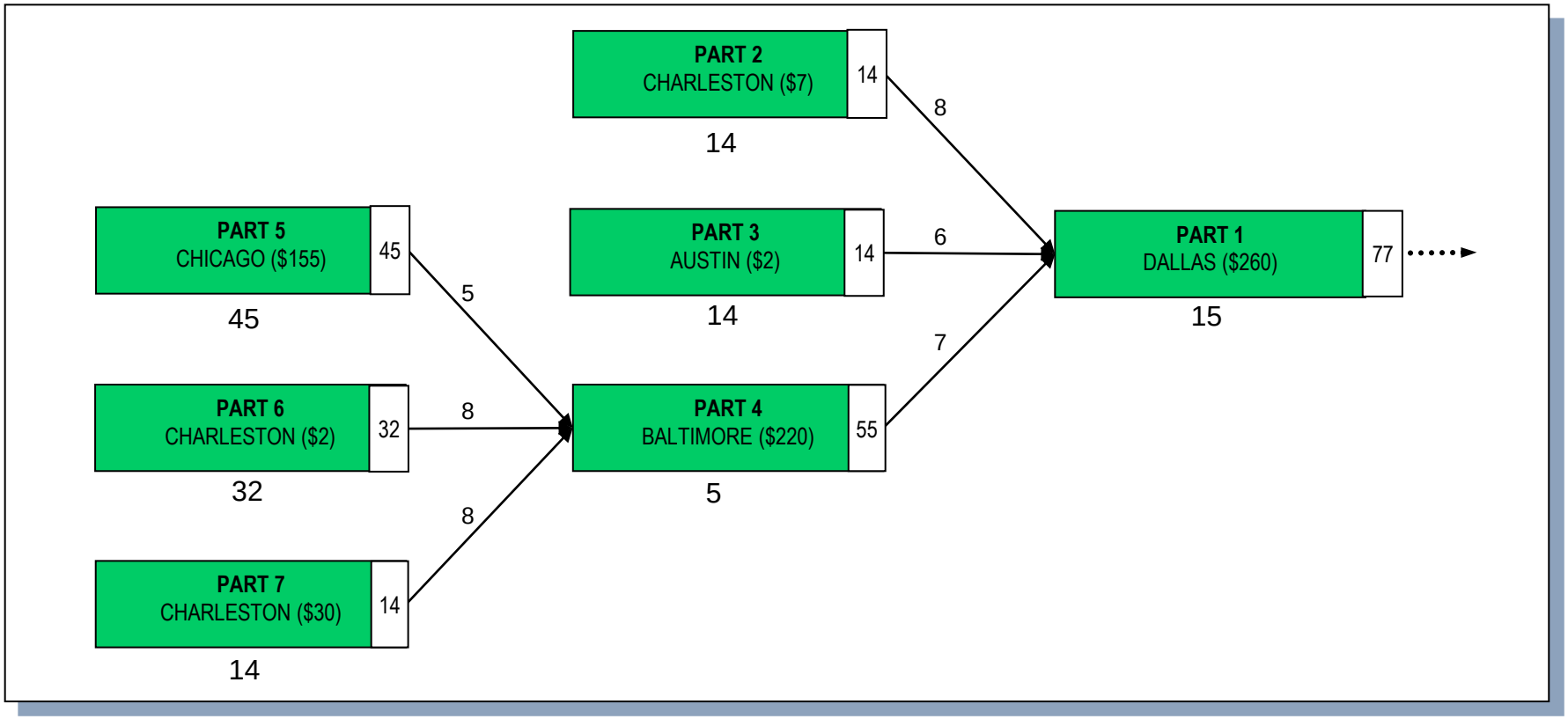
How to read the diagrams



Source: Adapted from Simchi-Levi, Chen, & Bramel, *The Logic of Logistics* (Springer, 2004)

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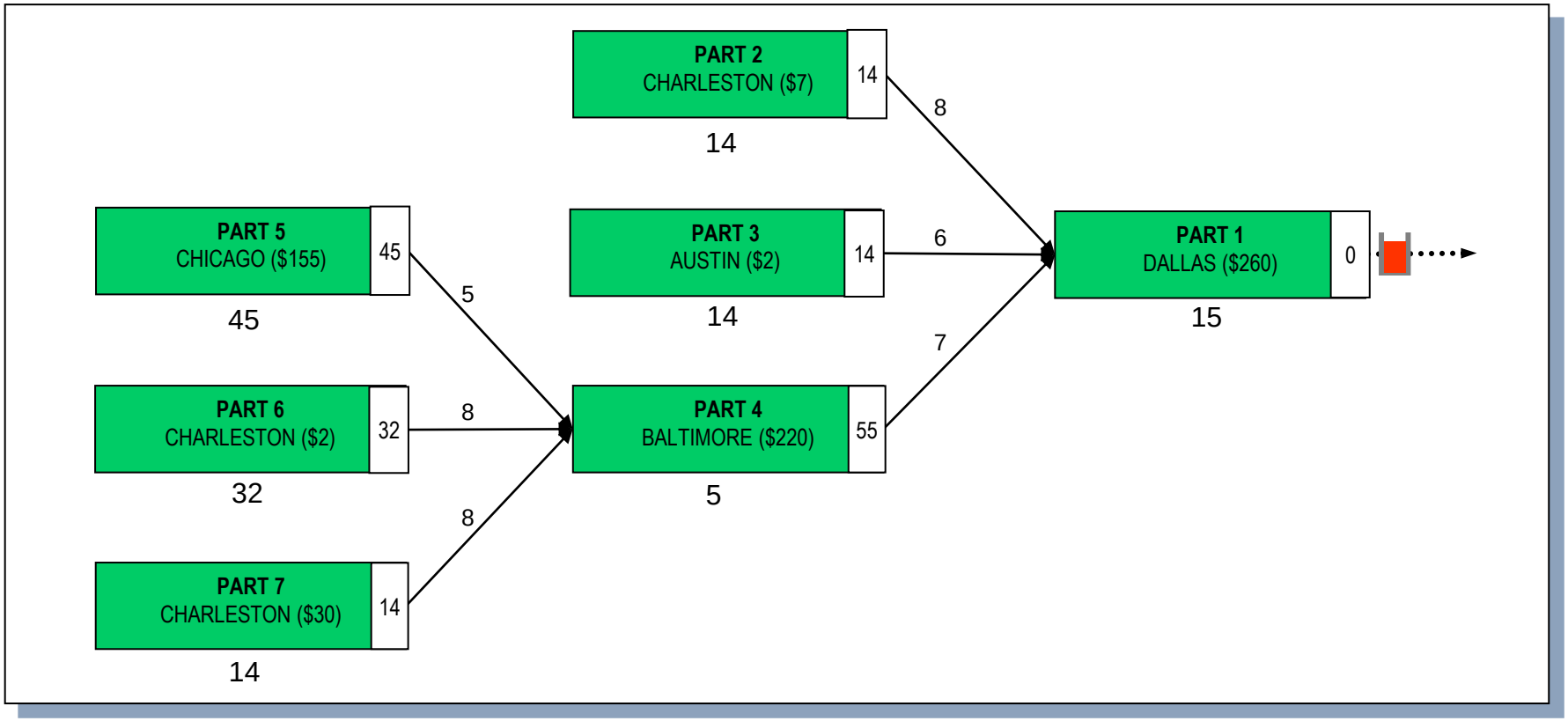
A pure "pull" system



- Produce to order
- Long committed service time (CST) to customer
- No inventory held in system

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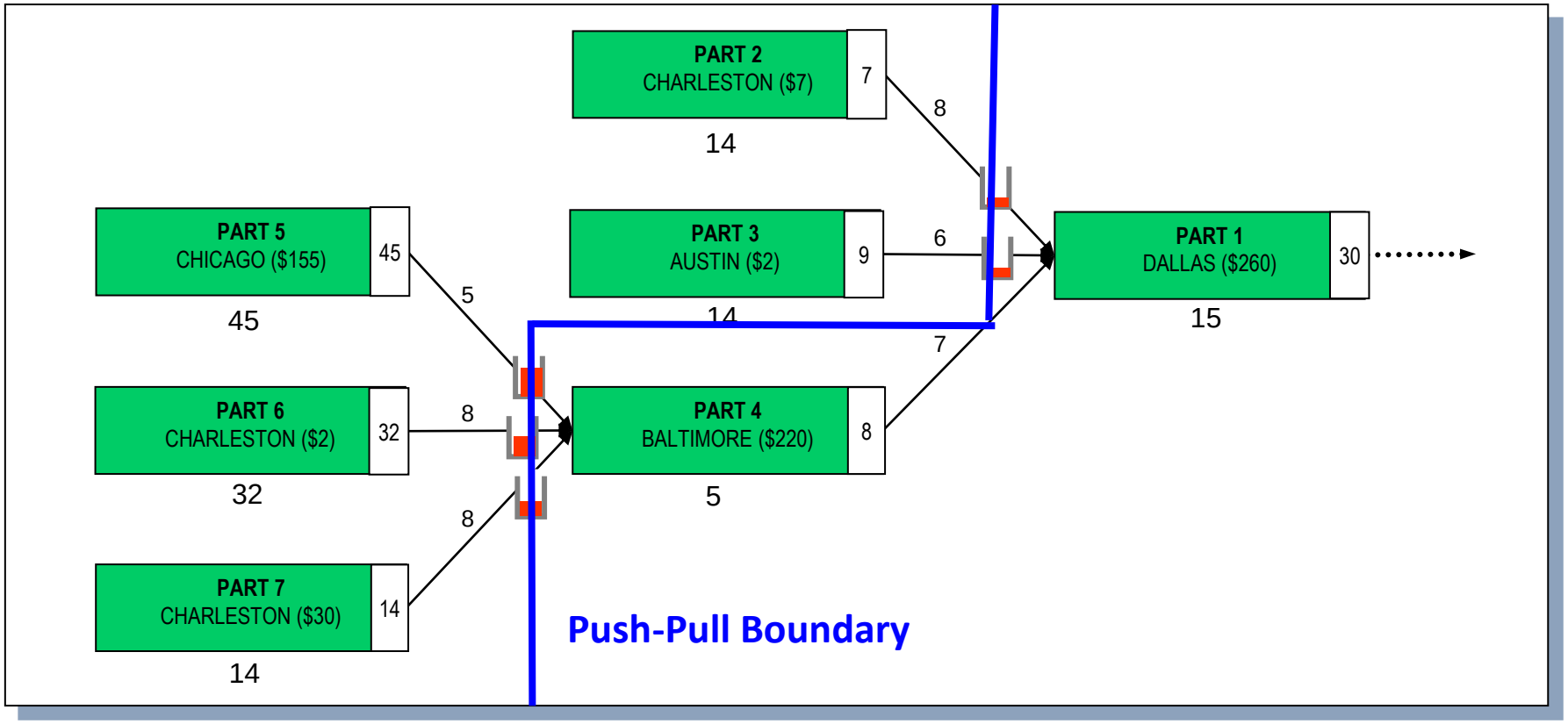
A pure "push" system



- Produce to forecast
- Zero CST to customer
- Hold lots of finished goods inventory

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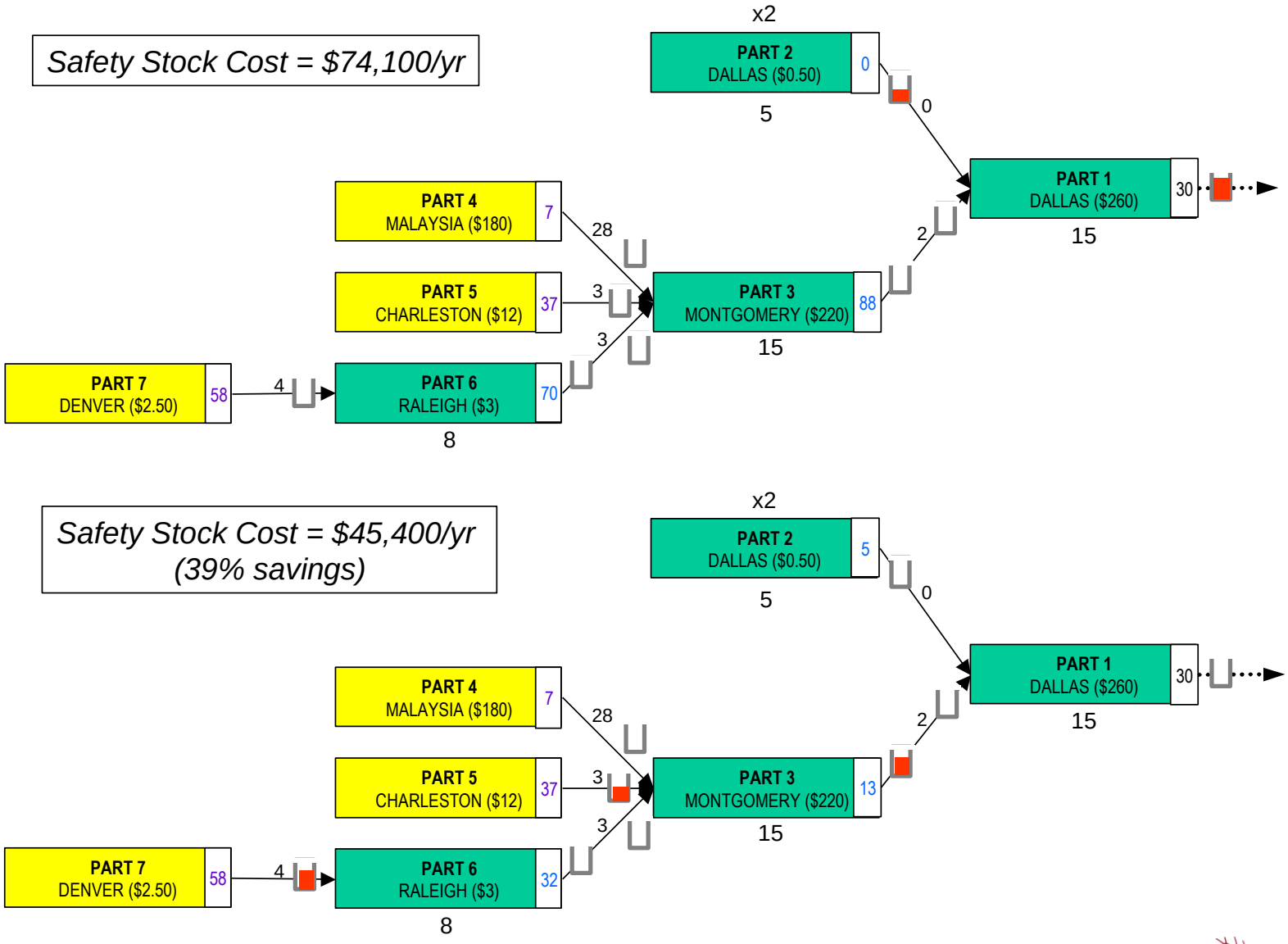
A hybrid "push-pull" system



- Part of system operated produce-to-stock, part produce-to-order
- Moderate lead time to customer

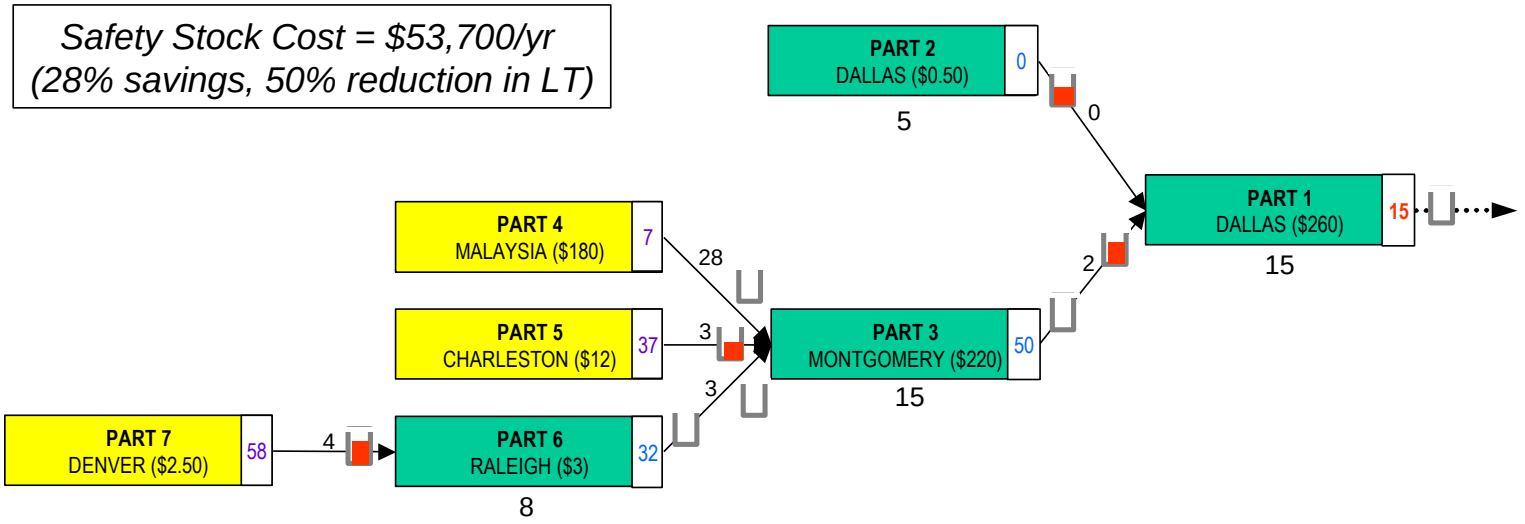
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Impact on cost and service level

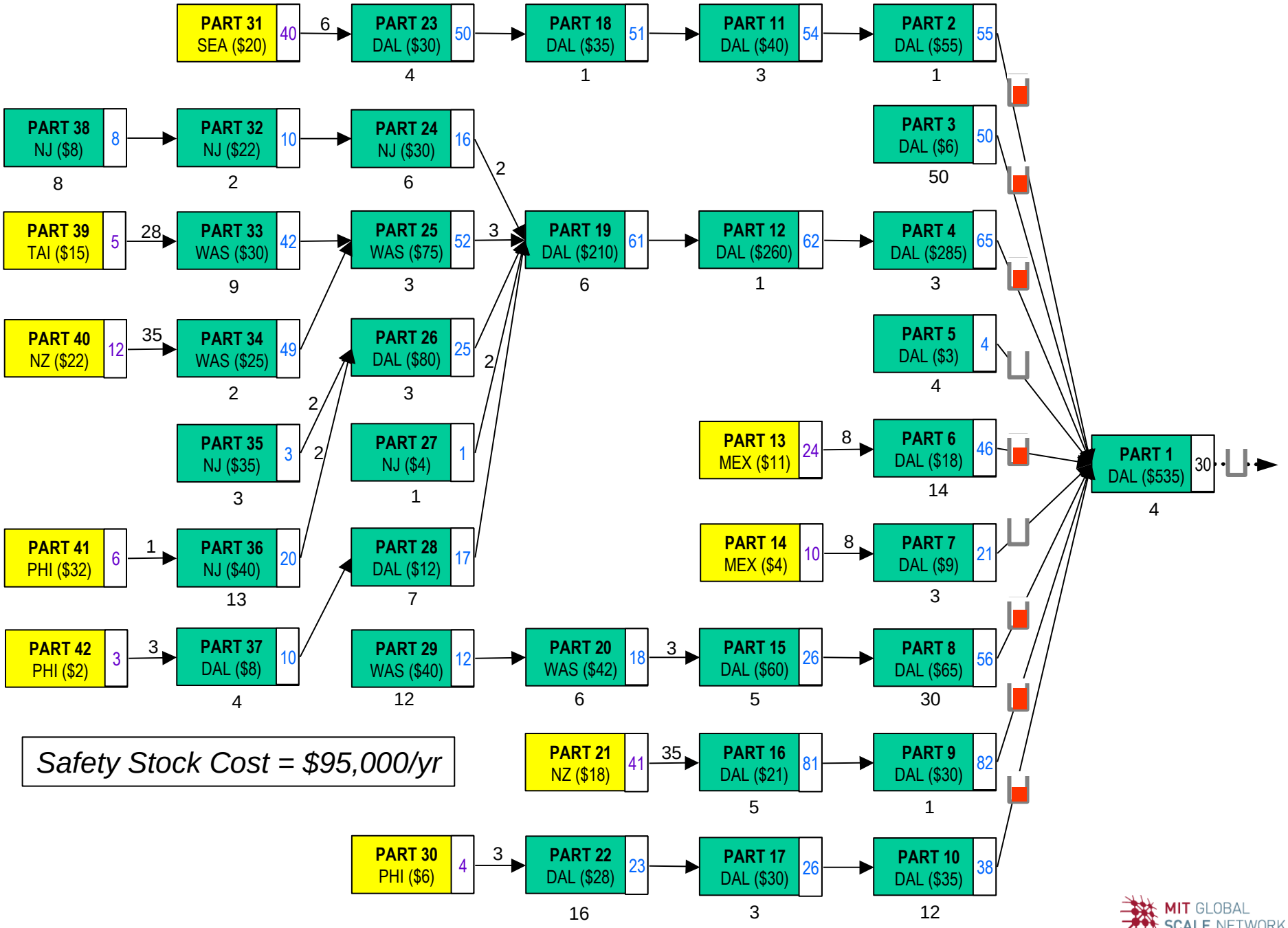


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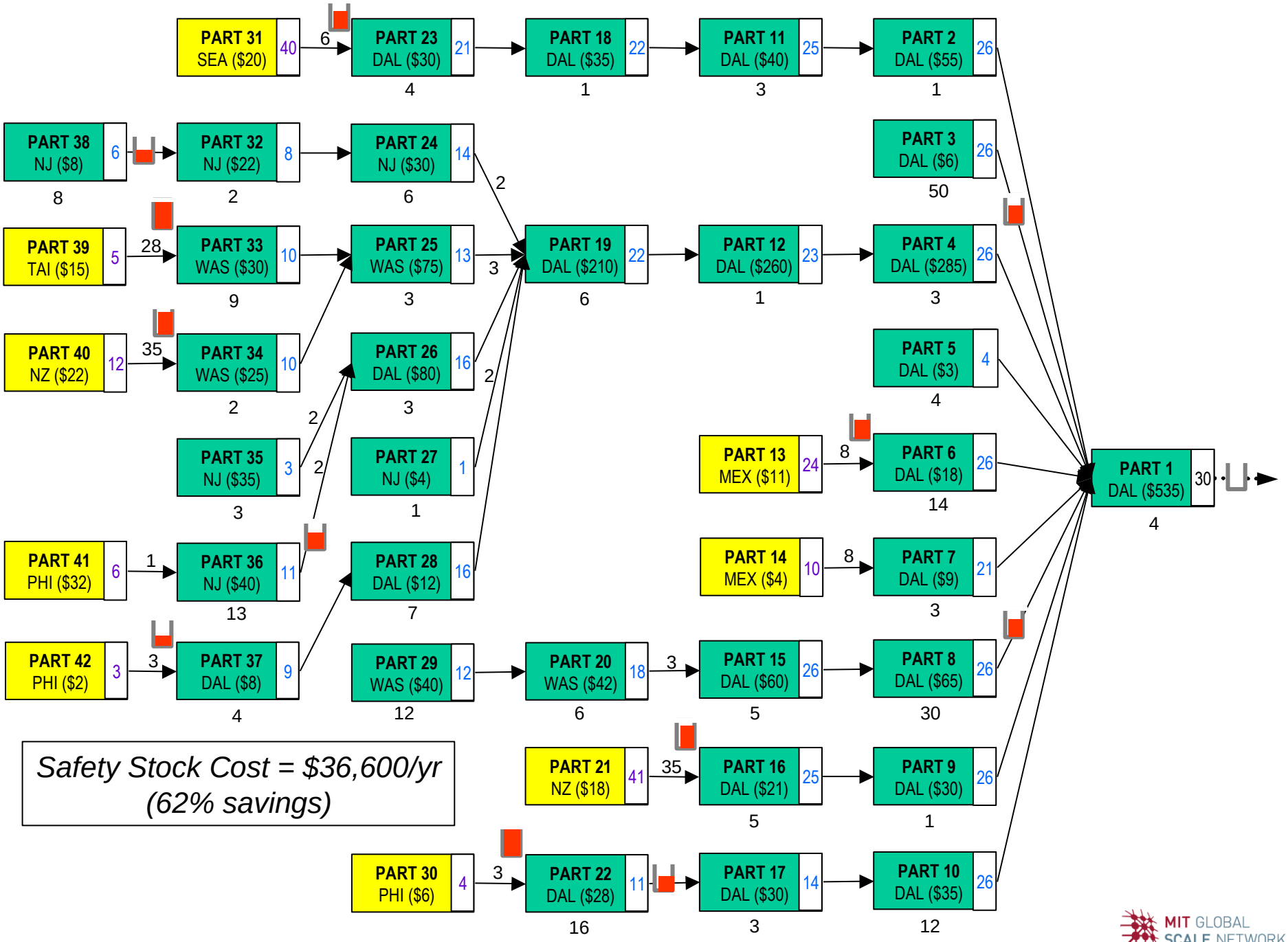
Impact on cost and service level



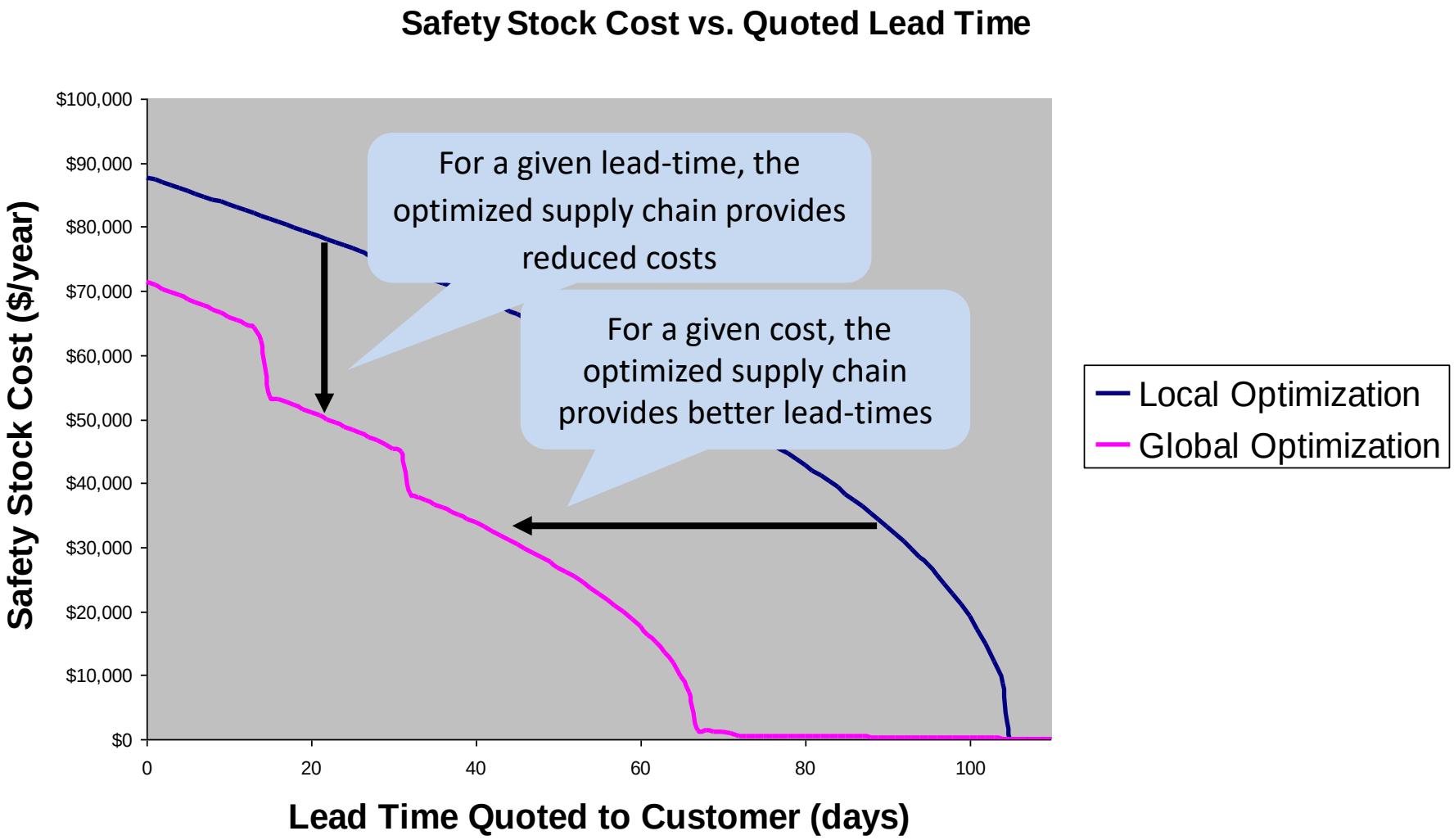
Scenario	Safety Stock Holding Cost ($/yr)	Lead Time to Customer (days)	Cycle Time (days)	Inventory Turns (turns/yr)
1: Baseline	$74,100	30	105	1.2
2: Optimization	$45,400	30	105	1.4
3: Shorten Lead Time	$53,700	15	105	1.3



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PART 31
SEA ($20)
40
6
PART 23
DAL ($30)
21
4
PART 18
DAL ($35)
22
1
PART 11
DAL ($40)
25
3
PART 2
DAL ($55)
26
1
PART 38
NJ ($8)
6
8
PART 32
NJ ($22)
8
2
PART 24
NJ ($30)
14
6
2
PART 3
DAL ($6)
26
50
PART 39
TAI ($15)
5
28
PART 33
WAS ($30)
10
9
PART 25
WAS ($75)
13
3
3
PART 19
DAL ($210)
22
6
PART 12
DAL ($260)
23
1
PART 4
DAL ($285)
26
3
PART 40
NZ ($22)
12
35
PART 34
WAS ($25)
10
2
PART 26
DAL ($80)
16
3
2
2
PART 5
DAL ($3)
4
4
PART 35
NJ ($35)
3
3
2
PART 27
NJ ($4)
1
1
PART 13
MEX ($11)
24
8
PART 6
DAL ($18)
26
14
PART 1
DAL ($535)
30
4
PART 41
PHI ($32)
6
1
PART 36
NJ ($40)
11
13
PART 28
DAL ($12)
16
7
PART 14
MEX ($4)
10
8
PART 7
DAL ($9)
21
3
PART 42
PHI ($2)
3
3
PART 37
DAL ($8)
9
4
PART 29
WAS ($40)
12
12
PART 20
WAS ($42)
18
6
3
PART 15
DAL ($60)
26
5
PART 8
DAL ($65)
26
30
PART 21
NZ ($18)
41
35
PART 16
DAL ($21)
25
5
PART 9
DAL ($30)
26
1
PART 30
PHI ($6)
4
3
PART 22
DAL ($28)
11
16
PART 17
DAL ($30)
14
3
PART 10
DAL ($35)
26
12
Safety Stock Cost = $36,600/yr
(62% savings)
MIT GLOBAL
SCALE NETWORK

From local to global optimization



Risk pooling: Summary

- Risk pooling is a theoretical (statistical) concept that suggests that demand variability is reduced if one can aggregate demand across products, location, time, etc.
- Redesign the supply chain, the production process, or the product to reduce uncertainty so that the firm is in a better position to mitigate its consequences in operational performance.
- Risk pooling strategies are less effective as demands across products, locations, time, etc. become more positively correlated.
- A little bit of risk pooling goes a long way.
- Risk pooling allows a firm to lower inventory and increase service simultaneously!

Readings

- Chapter 15 from Cachon & Terwiesch, 3rd ed.

Thank you!